Ariadne

Collaborative 3D Reconstruction and Egress Mapping
in Hostile, Communication-Constrained Environments

Abstract

After a structural collapse, the first hours determine whether trapped people are found and whether responders can reach them safely. The spaces that matter most are often ones humans cannot enter, because they are unstable, unlit, obstructed by rubble, and cut off from positioning and communication infrastructure. A small fleet of autonomous visual agents can enter such a space, but sending them in independently is not sufficient. Each agent has to decide what is worth observing, what is worth transmitting over a degraded link, and which of several apparently conflicting observations describe the same physical location.

Ariadne studies this as a single question: how should multiple autonomous visual agents collaboratively decide what to observe and how to reconcile their observations while constructing a consistent 3D world model under a shared mission budget? The system we propose to build couples a feature-based visual front-end, a from-scratch visual SLAM stack, bandwidth-aware multi-agent coordination, robust geometric consensus, active perception, and dense 3D reconstruction. Its output is both a spatial model for incident command and a continuously updated, risk-weighted egress route through that model, which is the thread that leads a responder in and back out again.

Perception and geometry will be built on OpenCV 5, whose restructured geometric modules and natively shipped learned feature stack make a hand-written SLAM system and a modern front-end compatible rather than mutually exclusive. Validation will proceed from public datasets with known ground truth to a physically grounded simulation of collapsed interiors. AWS will provide the reconstruction and evaluation compute, the Arm deployment path for the per-frame image workload, and the orchestration layer that runs the agent fleet.

1 The Problem

Consider a partial collapse following an earthquake. The structure is unstable, power is out, and the interior has become a set of narrow voids between concrete, furniture and debris. People may be trapped inside, and entering without a current understanding of the geometry exposes responders to avoidable risk. The survival window that search and rescue teams work against sets the operational frame: the objective is not a perfect map at some later time, but enough usable knowledge, quickly enough, to inform decisions while the window is still open.

The agents that enter operate under several simultaneous limits. Battery capacity is finite and cannot be replenished inside the structure. Wireless links are attenuated by rubble and reinforced concrete, so neither continuous streaming to a base station nor reliable peer-to-peer contact can be assumed. There is no dependable global positioning, and mission time is bounded by the response window itself. These constraints interact, which is part of what makes the coordination problem interesting: an observation that is cheap in bandwidth early in a mission may be unaffordable in energy later, and the agent best placed to resolve a particular uncertainty may be the wrong one to send if it is the only agent still holding a link to the rest of the team.

The environment also degrades visual estimation directly. Concrete and rubble present large regions of low texture, and structural elements repeat, so physically distinct locations can produce

nearly identical images. This perceptual aliasing is the failure mode that most threatens a multi-agent system, because place recognition is precisely the mechanism by which independently built maps are joined, and a confident false match propagates error into every trajectory the global optimiser subsequently tries to satisfy. Where power is absent and agents carry their own lighting, illumination is coupled to camera pose, so surface appearance changes as an agent moves even when its motion is estimated correctly.

Each agent therefore holds a partial and imperfect view. The team has to decide which missing observations are worth acquiring, who should acquire them, what to communicate, and which measurements to trust.

2 Research Question and Hypothesis

> **How should multiple autonomous visual agents collaboratively decide what to observe and how to reconcile their observations while constructing a consistent 3D world model under a shared mission budget?**

The question deliberately leaves the budget plural. Sensing, motion, energy, elapsed mission time and communication are all finite, they trade against one another, and their relative scarcity shifts during a mission. A policy that is right when bandwidth is the binding constraint is not the policy that is right when an agent is approaching its endurance limit.

Our hypothesis is that a team allocating observations and messages according to their expected contribution to the shared world model will produce a more accurate and more operationally useful result, for the same mission budget, than a team of agents each maximising its own local map coverage. Three coupled claims follow, and the evaluation is designed so that each can be isolated by ablation. Sensing and motion should be spent where uncertainty is consequential rather than where coverage is merely incomplete. Communication should be spent on the inter-agent relationships that most improve the structure of the joint estimation problem rather than on the visually strongest matches, which tend to be redundant with one another. Geometric consensus should remain stable under a high rate of false correspondences, since in self-similar rubble that rate is the expected operating condition rather than an unusual one.

3 Technical Approach

3.1 Visual front-end

The front-end will be geometric and feature-based rather than photometric. Direct methods rest on brightness constancy, which holds poorly in the target environment and fails outright once agents carry their own lighting, since the light source then moves with the camera and scene radiance becomes a function of pose. Committing to a feature-based front-end at the outset means the illumination work described in Section 7 extends the system rather than forcing it to be rebuilt.

The pipeline separates two tiers of cost. Frame-to-frame tracking uses sparse optical flow on keypoints carried from the previous keyframe, which keeps pose estimation running at frame rate on modest onboard hardware. Learned feature extraction and matching, which are considerably more expensive, are reserved for keyframes and for observations that may contribute to inter-agent registration. Besides bounding per-agent compute, this attaches communication to mapping events rather than to individual frames, which is what makes the bandwidth accounting in Section 3.3 tractable.

A restoration stage precedes feature extraction, covering denoising at low signal levels, illumination and vignette compensation, and contrast recovery under airborne dust. We intend to ablate this stage rather than assume it helps. Restoration that improves the appearance of an image while degrading the geometry recovered from it is a realistic outcome, and if we measure it we will report it.

3.2 Local mapping and SLAM

Each agent maintains its own trajectory and map before any attempt at reconciliation with the team. The local system is keyframe-based, with sparse tracking between keyframes, keyframe insertion on sufficient viewpoint change, triangulation of local landmarks, and windowed bundle adjustment over a small set of recent keyframes. Poses are optimised on the manifold using analytic Jacobians under a single fixed perturbation convention, and landmark blocks are eliminated by Schur complement so that computation concentrates on the camera variables.

Above the windowed estimator sits a pose graph, with odometry edges between consecutive keyframes and loop closures and inter-agent constraints added as they are verified. For pose graph optimisation, initialisation matters more than the choice of solver, so rotations will be initialised by chordal relaxation before translation and joint refinement, and the gauge freedom of the problem will be handled explicitly rather than left for the solver to encounter.

Because both the estimator and the simulator are being written from scratch, either can be wrong in a way that resembles a fault in the other. We will therefore keep an established optimisation library in the test suite as a correctness oracle, solving the same pose graph both ways and reporting agreement, while keeping it out of the runtime path.

3.3 Communication under a bandwidth budget

The dominant communication cost in collaborative SLAM is not the exchange of pose estimates. It is the descriptor and keyframe traffic required to discover inter-agent relationships in the first place, which typically exceeds backend optimiser traffic by one to two orders of magnitude. Ariadne will account for both categories separately and concentrate effort on the larger one.

Exchange is hierarchical. Agents broadcast only compact global descriptors for each keyframe, which is enough for a peer to identify that a useful relationship may exist. Since these descriptors are broadcast per keyframe, descriptor dimensionality becomes a first-order term in total mission bandwidth, and the choice of place-recognition embedding is as much a bandwidth decision as an accuracy one. Candidate relationships are then prioritised under an explicit budget according to their contribution to the connectivity of the joint pose graph, using information already available from the broadcast stage and requiring no additional communication to evaluate. This favours links that connect weakly related regions of the map over links that are visually strong but redundant with links already selected. Expensive local descriptors and the geometric evidence needed for verification are requested only for candidates that survive prioritisation, and where several selected candidates share a keyframe, its features are transmitted once. Bookkeeping ensures nothing is sent to a peer twice, and reference frames converge as agents meet, without requiring the whole team to be in contact simultaneously.

The backend follows the same reasoning. Rather than iterating a distributed optimiser across an unreliable network, an elected broker collects compact local pose graphs when agents are in contact, runs a robust global optimisation, and returns updated estimates. Iterating an optimiser over the link costs substantially more bytes than shipping the graph once, and it also makes correctness depend on stateful synchronisation, so a lost packet can silently move the

team toward a different solution. Under the broker architecture a lost packet costs delay or information rather than consistency.

3.4 Robust geometric consensus

An incorrect inter-agent loop closure is more damaging than a missing one, because the global optimiser will deform otherwise correct trajectories in order to satisfy it. Inter-agent loop closures are therefore treated as hypotheses and filtered in successive layers. Two-view geometric verification rejects candidates whose correspondences are not consistent with a plausible relative pose. A pairwise consistency check then tests whether composing two candidate closures with the intervening odometry returns close to identity, which removes candidates that are individually plausible but mutually incompatible. The primary mechanism is graduated non-convexity in the optimiser itself, which begins from a convex surrogate and progressively sharpens toward the true robust objective while reweighting measurements in closed form, so the existing weighted least-squares solver is reused unchanged. Robust losses apply to loop-closure edges only, since odometry edges are not the source of outliers.

For selected pose graph instances we also intend to compute an a-posteriori optimality certificate for the rotation component. The certificate is not required for normal operation; it provides an independent check of whether the merged multi-agent solution is globally optimal under the estimator's assumptions, which is otherwise difficult to establish for a non-convex problem.

Optimisation and agreement are kept distinct throughout. Geometric estimation decides which measurements are consistent with the world model, while exact consensus protocols are used only where exact agreement is genuinely the objective, such as broker election, configuration, and task ownership.

3.5 Active perception and task allocation

Agents do not simply explore until the accessible space has been photographed. The system maintains an estimate of where the current model is uncertain, incomplete or poorly connected, and each agent proposes candidate actions against it: moving toward an unexplored region, revisiting an ambiguous one, approaching a peer to establish a useful cross-agent observation, or collecting additional views of a geometrically important surface. Candidates are scored as

$$U(a) = I(a) - \lambda_d C_{\text{motion}}(a) - \lambda_b C_{\text{bandwidth}}(a) - \lambda_e C_{\text{energy}}(a) - \lambda_t C_{\text{time}}(a),$$

where $I(a)$ is the expected contribution of the action to the shared model and the remaining terms are its operational costs. The weights are not fixed for the duration of a mission. As the response window narrows or an agent approaches its endurance limit, the value of further exploration relative to consolidating what is already known changes, and the allocation should change with it.

Because each agent's best action depends on what its peers are already doing, this is an allocation problem rather than a set of independent view planners. Agents share a lightweight view of claimed objectives and current coverage, which deconflicts trajectories and prevents the team from spending twice to acquire the same information.

3.6 Human authority and risk gating

Some volumes remain structurally ambiguous even after reconstruction. Where a proposed action would enter a region whose estimated risk exceeds the autonomous operating envelope,

the agent requests approval from incident command rather than proceeding. The operator is not asked to pilot the fleet; they receive a structured decision containing the supporting evidence, the expected gain, and the reason the action was escalated. If the link to incident command is lost, agents fall back to a conservative policy, continuing work that lies within the pre-approved envelope while deferring anything requiring external approval.

3.7 Dense reconstruction

The final spatial representation will be a 3D Gaussian splat built from validated keyframes and optimised camera geometry, so that incident command can inspect the recovered space from viewpoints that were never directly observed and can see what the agents have actually learned rather than an abstract occupancy structure. Agents build local submaps and exchange only what is required to align and merge them, so dense representations are not streamed continuously.

Two aspects of the merge are scoped from the outset. Agents run independent exposure control and see the same surface under different lighting, so a per-keyframe colour correction will be estimated alongside the reconstruction to keep photometric differences from appearing as seams in the shared model. Overlap between submaps also has to be resolved rather than accumulated, and the specific mechanism is an open implementation decision we expect to settle empirically, starting from a simple geometric resolution of overlapping regions and measuring what a more sophisticated alternative buys before adopting one.

The dense layer remains downstream of the geometric one throughout. A convincing render is not treated as evidence of a correct map unless the camera geometry and inter-agent registration underneath it have been validated independently.

4 The Ariadne Thread

For incident command the operative question is usually not what the building looks like, but how to get a person to a particular location and back out. Ariadne therefore treats the egress route as a first-class output rather than a downstream visualisation.

The reconstruction is converted into a spatial cost field combining traversability, clearance, reconstruction uncertainty and estimated structural risk, and a graph search over that field produces routes between locations of interest and a validated exit. Routes are updated continuously as observations arrive. A corridor that becomes well observed falls in cost, a newly revealed obstruction invalidates and reroutes, and a path through poorly observed volume carries lower confidence than one supported by several consistent observations. This puts the uncertainty machinery to operational use, and it produces metrics a non-specialist can interpret: how long the team takes to establish a first valid route, and whether the recommended route is actually traversable in ground truth.

5 OpenCV 5 in the Stack

OpenCV 5 is load-bearing at both the pixel level and the geometry level rather than serving as a preprocessing dependency. Two properties of the 5.x line matter specifically. The first is the reorganisation of the old calibration module into separate calibration, geometry, stereo and point cloud modules, which is where most of the geometric work in this project lives. The second is that OpenCV 5 ships a modern learned feature stack natively and under permissive licensing, which removes the usual tension between writing a SLAM system yourself and using a current

front-end inside it.

Stage	Module	Use
Restoration	`photo`, `imgproc`	denoising, illumination compensation, contrast recovery
Tracking	`video`	pyramidal optical flow between keyframes
Keyframe features	`features`	`cv::XFeat` (64-D), `cv::ALIKED`, `cv::DISK`
Matching and retrieval	`features`	`cv::LightGlueMatcher`, `cv::ANNIndex`
Two-view geometry	`geometry`	essential matrix, `solvePnPRansac`, triangulation
Calibration	`calib`	intrinsics, multi-camera rigs
Short-baseline depth	`stereo`	disparity in confined spaces
Point clouds	`ptcloud`	registration, merging, file I/O
Evaluation	`quality`	image metrics on held-out views

Two capabilities are used directly rather than reimplemented. USAC is the default robust estimation backend across homography, essential matrix and perspective-n-point estimation in OpenCV 5, which provides modern sample consensus without a hand-written sampler. And the compact 64-D descriptors produced by `cv::XFeat` are the smallest of the available learned extractors, which matters here because those descriptors are transmitted between agents.

Everything above these primitives will be written from scratch: keyframe management, the local map representation, bundle adjustment, the pose graph and robust backend, the inter-agent protocol, task allocation, byte accounting, and the egress layer. Learned components are used through OpenCV's ONNX path rather than reimplemented, since the time is better spent on the estimator. Component licences are checked individually, and we will use only permissively licensed weights and reference implementations.

6 Agentic Orchestration

The coordination layer sits on top of deterministic estimators and does not replace them. Pose, depth, registration, uncertainty and coverage utility are computed classically and are measurable, while the orchestration layer reasons over those outputs about allocation, negotiation, budget trade-offs and exception handling.

OpenCV 5 operations, including restoration, feature extraction, geometric verification and registration, will be exposed as MCP-compatible tools so that the orchestrator invokes vision work through one standardised and loggable interface. This gives a clean record of the property that matters operationally, which is a visual result changing what the system does next, whether that is a movement, a request to a peer, or an escalation to a human operator. Every such decision is logged with the vision-derived inputs that produced it, and the orchestration layer will itself be ablated against a purely classical planner. If classical planning performs as well, we will say so.

7 Scope and Staging

The project is staged so that a working end-to-end system exists early and each subsequent layer is added to something already validated.

The minimum viable system is the core of the research question: several agents running local visual SLAM, discovering and verifying inter-agent relationships under an explicit bandwidth

budget, merging into a consistent shared model through the robust backend, reconstructing densely, and producing an egress route. It is evaluated first in conditions where texture-poor geometry and perceptual aliasing are present but illumination is conventional, which is enough to exercise coordination, consensus and allocation without leaving the perception stack and the coordination stack uncertain at the same time.

Self-carried illumination is the first extension. It adds camera-attached lighting with inverse-square falloff and cast shadows to the simulator, and the corresponding illumination compensation and low-signal restoration to the front-end. The front-end is designed with this case in mind from the beginning, so the extension is a matter of adding modelling and restoration rather than changing the architecture.

Smoke, flame artifacts and thermal fusion are a stretch goal, attempted only once the base system is complete and evaluated. Denser participating media extends the dust model rather than replacing it, and a thermal channel is valuable mainly as a witness channel when the visible spectrum fails entirely and as a route to victim detection. If time runs out, this is a documented roadmap item rather than a half-built subsystem.

8 Validation

Validation begins on public data. Indoor and subterranean datasets with known trajectories and geometry will be used to validate the front-end, local SLAM, relative pose estimation and reconstruction independently of any simulator. This ordering is deliberate and is the single practice most likely to protect the schedule, because a from-scratch estimator and a from-scratch simulator can both be wrong while remaining mutually consistent, and separating the two later is expensive. Where public data supports a claim we will make it, and where it does not, particularly for rubble and for occluded casualties, we will state the limitation rather than extrapolate.

The simulator then supplies conditions that no public dataset provides in controlled form: a collapsed interior with narrow passages and repeated structural elements, tunable dust density, sensor noise at low signal levels, communication attenuation and packet loss, and, at the illumination stage, camera-mounted lighting with physically plausible falloff and shadowing. Clean ground truth is retained beneath every degradation so that tracking and reconstruction error can be measured against the undegraded scene, and the simulator also supplies ground-truth traversability so that a proposed egress route can be checked rather than merely rendered. Fidelity effort goes into the properties that affect the algorithms, since a physically faithful illumination and noise model is worth more here than a visually impressive renderer.

Failure injection is part of the experimental design. The suite will introduce network partitions and intermittent links, agents lost mid-mission, an agent reporting degraded poses after an impact, tracking loss in featureless regions, false inter-agent matches arising from perceptual aliasing, conflicting local maps, and observations arriving late or out of order. The question is not whether Ariadne avoids these failures, but whether it detects them, contains their propagation, and degrades predictably when its assumptions stop holding.

9 Evaluation

The primary relationship under study is world-model quality against mission budget spent, where budget covers bytes exchanged, distance travelled, an energy proxy and elapsed mission time. We will compare independent exploration, coverage-driven coordination, and Ariadne's uncertainty-

and connectivity-aware policy, measuring trajectory and registration accuracy, reconstruction completeness and visual quality, consistency of the shared model, communication volume by category, time to a usable shared map, time to first valid egress route, egress validity against ground truth, and the rate at which each of these degrades under communication loss and visual corruption.

Bytes are counted at a single serialisation boundary and tagged by category, so the accounting reflects what actually crossed the link rather than what the algorithm considered a message. The headline figure is error against cumulative budget spent, one curve per coordination policy, and we will report the budget required to reach a quality threshold rather than the budget consumed at convergence, since the latter rewards stopping early.

Four ablations test the hypothesis directly. Observation selection is replaced by random or coverage-driven choice. Connectivity-aware communication prioritisation is removed, so observations are exchanged without regard to their contribution to the joint estimation problem. Robust consensus is disabled, exposing the effect of false loop closures. The coordination layer is replaced by independent exploration. If collaborative decision-making earns its cost, the advantage should appear as better geometry and more reliable routes for the same budget rather than as a better-looking render.

10 AWS Deployment and Compute

Individual agents are designed around lightweight local perception and mapping, so the cloud carries three things: compute, the Arm path for the highest-volume image workload, and orchestration of the agent fleet and its deployment.

Compute is the largest of the three. Dense reconstruction and the evaluation programme both run off-agent, and the evaluation programme is the more demanding, since the central claims rest on comparing coordination policies across repeated controlled missions while varying agent count, link quality, degradation severity and policy. Running those sweeps in parallel on EC2, with mission artifacts and results in S3, is what turns the comparison into a measured result rather than a single demonstration run.

The per-frame restoration and feature preparation stage is the highest-volume workload in the system, since it runs on every frame from every agent, and it consists largely of the image processing primitives that the Cloud-Optimized OpenCV Library accelerates on Graviton. We intend to deploy that stage on the COOL AMI, which builds on OpenCV 5 and requires no code changes, and to benchmark it against a defensible baseline: the same pipeline and inputs on the same instance type, COOL against a stock OpenCV 5 build, reporting throughput, utilisation and cost per frame, then sweeping agent count to answer the operationally meaningful question of how many agents one instance can serve and at what cost per mission. The free trial covers the initial characterisation, and the Arm and x86 build paths will both be pinned and reproducible.

Orchestration covers running the agent fleet, simulated or physical, alongside the coordination layer and the incident-command console, and deploying the same containers across both architectures. The console shows the live world model, agent positions and camera feeds with restored output alongside raw, coverage and uncertainty, remaining budgets, the inter-agent negotiation record, the growing reconstruction, the current egress thread, and the approval control for risk-gated actions. In operational mode agents decide locally and use the network only for what coordination requires, so the cloud path is a compute and orchestration tier rather than a dependency for correctness.

Requested credits would go to the evaluation fleet, the reconstruction tier, the Graviton bench-

mark, and hosting the console. Cost is dominated by the number of repetitions in the evaluation sweeps rather than by any single large training job.

11 Responsible Use

Ariadne is built for environments whose defining property is that a human cannot safely enter them, and its beneficiary is the responder who would otherwise have to. The system requires geometry, traversability, uncertainty and routes. It does not require person identification or persistent identity tracking, and will not implement them.

The dual-use concern is worth stating plainly, since multi-agent visual mapping of building interiors could be misapplied to surveillance or to unauthorised reconnaissance. We address it structurally rather than by assertion, by restricting the representation to geometry and traversability, keeping high-consequence actions behind explicit human approval, and exposing uncertainty rather than presenting an estimated route as a guarantee of structural safety. Ariadne is a decision-support model of an environment. It does not replace professional structural assessment or human command authority.

12 Conclusion

A collapsed structure is a demanding case for visual intelligence. It is only partially observable, positioning is unavailable, communication is unreliable, surfaces are poor in texture, and physically different locations produce similar visual evidence. The response to that is not simply more robots, because a team of agents with excellent local perception can still fail collectively by observing the same region, exhausting their budgets without resolving a decisive uncertainty, saturating a weak link with redundant descriptors, or accepting a plausible correspondence that corrupts the shared map.

Ariadne treats those failures as one connected problem, closing the loop from observation through estimation, uncertainty, negotiation and motion back to observation, under budgets that are all finite at once. The output is deliberately concrete: a globally registered photorealistic model of a space nobody could safely enter, and a continuously validated route through it that leads a responder in and back out again.